UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	:	COMPUTER SECURITY
COURSE CODE	:	CSC662
EXAMINATION	:	OCTOBER 2026 (MOCK)
TIME	:	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions)
   PART B (2 Questions)

2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.

3. Do not bring any material into the examination room unless permission is given by the invigilator.

4. Please check to make sure that this examination pack consists of:

   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty

5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages

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PART A (80 MARKS)

QUESTION 1

a) List TWO (2) typical locations where IDS sensors should be placed to defend valuable assets.

(2 marks)

b) Explain six preventive network security mechanisms, focusing on firewalls, proxy servers, segmentation, VPNs, ZTNA, and IDS/IPS, describing how each protects an organization's network.

(6 marks)

QUESTION 2

a) Explain the fundamental mechanism of a substitution cipher and identify the cryptographic principle it corresponds to.

(2 marks)

b) Decrypt the following ciphertext using the Vigenere cipher with the key CIPHER. Use the standard 26 x 26 Vigenere tableau, writing the key over the message and repeating it as necessary.
UMRBVVNWVPREQE

(6 marks)

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Vigenere tableau (A = 0)

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

QUESTION 3

a) List TWO (2) key sources or characteristics of software insecurity as described in the lecture.

(2 marks)

b) Explain the key security mechanisms and architectural features of Windows operating systems, including their evolution, management structures, and potential vulnerabilities.

(6 marks)

QUESTION 4

a) State TWO (2) characteristics of Trojan horses and computer worms as described in the lecture slides.

(2 marks)

b) Discuss six (6) distinct malware attack techniques or propagation methods described in the lecture slides.

(6 marks)

QUESTION 5

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a) Explain the distinction between morals and laws in the context of computer security ethics.

(2 marks)

b) State six principles from the ACM Code of Ethics that guide a computing professional's conduct regarding society, honesty, fairness, intellectual property, privacy, and confidentiality.

(6 marks)

QUESTION 6

a) Explain the two main stages of the biometric authentication process and list one key step involved in each stage.

(2 marks)

b) Explain the definition of a biometric and its two main categories, providing specific examples for each category and one disadvantage associated with fingerprint recognition.

(6 marks)

QUESTION 7

a) State the primary security focus of the Bell-La Padula (BLP) model and the Biba model.

(2 marks)

b) Describe the characteristics and trade-offs of Access Control Matrix (ACM), Access Control List (ACL), and capability structures in access control systems.

(6 marks)

QUESTION 8

a) Explain two ethical responsibilities of a computing professional regarding the evaluation and public understanding of technology impacts.

(2 marks)

b) Refer to your answer in (a). Discuss how professional codes of ethics and documented behavioral standards guide organizational responsibility regarding privacy and conduct.

(6 marks)

QUESTION 9

a) Explain how the threat classes of Disclosure and Usurpation specifically impact the Confidentiality and Availability objectives of the CIA triad.

(2 marks)

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b) State six (6) distinct characteristics or impacts of the Samy worm and Cross-Site Scripting (XSS) attacks as described in the lecture slides.

(6 marks)

QUESTION 10

a) State TWO (2) limitations of Misuse Detection Systems in identifying attacks.

(2 marks)

b) Discuss six (6) distinct software threats or malware propagation techniques described in the lecture slides.

(6 marks)

PART B (20 MARKS)

QUESTION 1

A security architect is designing a platform that relies on hardware-based trust. The design specifies the use of a Trusted Platform Module (TPM) to establish a chain of trust from boot time through to application execution, ensuring the integrity of the system.

a) State five key characteristics or functions of the TPM and its role in establishing a chain of trust.

(5 marks)

b) Discuss the components of the Next Generation Secure Computing Base (NGSCB) and their roles in establishing hardware-based trust within the scenario.

(5 marks)

QUESTION 2

A university IT department is reviewing its user authentication protocols to enhance security. The team focuses on strengthening password management practices and implementing controls to prevent unauthorized access through compromised credentials.

a) Explain five specific password security controls or policies that can be applied to protect user accounts and system integrity.

(5 marks)

b) In the scenario, explain five key aspects of Kerberos and Single Sign-On mechanisms that support secure authentication in distributed environments.

(5 marks)

END OF QUESTION PAPER

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